



Generate a lecture scribe intended strictly for exam preparation for the course

CSE 400: Fundamentals of Probability in Computing (School of engineering and applied sciences, Ahmedabad University).

Strictly use only the provided context, which consists of:

- Marginal PMF
 - Definition
 - Marginal PMF from a Joint PMF Table
 - Solved Example
- Correlation
- Covariance
 - Definition and interpretation
 - Algebraic form and Properties
 - Solved examples
 - Pearson's Coefficient
- Joint Gaussian RV
 - Definition and intuition
 - Algebraic form and interpretation
 - Coding Exercise

Do not introduce any concepts, results, derivations, explanations, interpretations, or examples that do not appear in the provided context.

While writing the lecture scribe, reason step by step and include the reasoning explicitly in the scribe for all of the following only when they appear in the context:

- Marginal PMF - Definition
- Marginal PMF - Examples
- Correlation - Definition
- Covariance - Definition and Interpretation
- Covariance - Algebraic Form and Properties
- Covariance - Examples
- Covariance - Pearson's Coefficient
- Jointly Gaussian RVs - Definition
- Jointly Gaussian RVs - Algebraic form and interpretation

Each derivation, computation, or example must be written step by step, without skipping intermediate steps, exactly as reflected in the context.

Maintain the authenticity of mathematical equations. Organize the scribe using clear section headings and logical flow, so that a student can rely on this document alone for closed-notes exam revision.

Do not:

- Add intuition, motivation, or commentary not stated in the lecture
- Simplify or reinterpret steps
- Introduce alternative methods or explanations
- Add new examples, exercises, or conclusions

Important Note - Strictly follow the instructions as mentioned above.

Below is a **lecture scribe strictly derived from the provided lecture slides**.

All definitions, equations, and example steps are reproduced **exactly from the lecture content** and organized for **closed-notes exam preparation**.

No additional concepts or explanations beyond the lecture have been introduced.

The material is sourced from the uploaded lecture file.

CSE 400 — Fundamentals of Probability in Computing

Lecture 18 Scribe

Marginal PMF, Correlation, Covariance, Joint Gaussian Random Variables

1. Marginal PMF

1.1 Definition

For discrete random variables X and Y , the **joint PMF** is

$$p_{XY}(x, y) = P(X = x, Y = y)$$

The **marginal PMFs** are obtained by summing over the other variable:

$$p_X(x) = \sum_y p_{XY}(x, y)$$

$$p_Y(y) = \sum_x p_{XY}(x, y)$$

Key Idea

The marginal PMF of one variable is obtained by **summing the joint PMF over all possible values of the other variable**.

1.2 Marginal PMF from a Joint PMF Table

Row sums give $p_X(x)$, and column sums give $p_Y(y)$.

$$p_X(x_i) = \sum_j p_{ij}$$

$$p_Y(y_j) = \sum_i p_{ij}$$

Normalization conditions:

$$\sum_i p_X(x_i) = 1$$

$$\sum_j p_Y(y_j) = 1$$

(Shown in the joint PMF table illustration on page 4 of the lecture slides.)

1.3 Example

Let the joint PMF be

$p_{XY}(x, y)$	$y = 0$	$y = 1$
$x = 0$	1/8	1/4
$x = 1$	3/8	1/4

Step-by-step computation

Compute $p_X(0)$

$$\begin{aligned}
 p_X(0) &= \frac{1}{8} + \frac{1}{4} \\
 &= \frac{1}{8} + \frac{2}{8} \\
 &= \frac{3}{8}
 \end{aligned}$$

Compute $p_X(1)$

$$\begin{aligned}
 p_X(1) &= \frac{3}{8} + \frac{1}{4} \\
 &= \frac{3}{8} + \frac{2}{8} \\
 &= \frac{5}{8}
 \end{aligned}$$

Compute $p_Y(0)$

$$\begin{aligned}
 p_Y(0) &= \frac{1}{8} + \frac{3}{8} \\
 &= \frac{4}{8} \\
 &= \frac{1}{2}
 \end{aligned}$$

Compute $p_Y(1)$

$$\begin{aligned}
 p_Y(1) &= \frac{1}{4} + \frac{1}{4} \\
 &= \frac{1}{2}
 \end{aligned}$$

Final Marginal PMFs

$$p_X(x) = \begin{cases} 3/8 & x = 0 \\ 5/8 & x = 1 \end{cases}$$

$$p_Y(y) = \begin{cases} 1/2 & y = 0 \\ 1/2 & y = 1 \end{cases}$$

2. Correlation

2.1 Definition

A basic way to measure joint behavior is through the **product moment**.

In the continuous case,

$$R_{XY} = E[XY]$$

It tells us the **average value of the product** XY .

This is often called a **raw correlation measure**.

2.2 Properties

$$R_{XY} = E[XY]$$

- Depends on the **scale of X and Y**
- Computed **about the origin**, not about the means

Therefore it mixes together

- mean values
- spread
- dependence

If

$$R_{XY} = 0$$

then the variables are **orthogonal about the origin**.

To remove the effect of the means, we **center the variables first**, which leads to **covariance**.

3. Covariance

3.1 Definition and Interpretation

Covariance is defined as

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

where

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Interpretation

First subtract the mean from each variable.

Then measure how the **centered quantities vary together**.

Covariance captures whether deviations from the means occur

- in the **same direction**
 - or in **opposite directions**.
-

Sign Interpretation

- $\text{Cov}(X, Y) > 0 \rightarrow$ Positive covariance
- $\text{Cov}(X, Y) < 0 \rightarrow$ Negative covariance
- $\text{Cov}(X, Y) = 0 \rightarrow$ Zero covariance

Covariance tells the **direction of joint linear variation** between X and Y .

However, the **magnitude depends on units** and cannot measure the strength of the relationship.

3.2 Algebraic Form

Start with

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

Expand the product:

$$= E[XY - X\mu_Y - \mu_X Y + \mu_X \mu_Y]$$

Apply linearity of expectation:

$$= E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y$$

Since

$$\mu_X = E[X], \quad \mu_Y = E[Y]$$

Therefore

$$\text{Cov}(X, Y) = E[XY] - \mu_X \mu_Y$$

or

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

(All algebraic steps appear exactly in the derivation on page 10.)

3.3 Properties of Covariance

1. Symmetry

$$\text{Cov}(X, Y) = \text{Cov}(Y, X)$$

2. Variance as a Special Case

$$\text{Cov}(X, X) = \text{Var}(X)$$

3. Linearity with Respect to Constants

$$\text{Cov}(aX + b, cY + d) = ac \text{Cov}(X, Y)$$

4. Independence Implies Zero Covariance

If

$$X \perp Y$$

then

$$\text{Cov}(X, Y) = 0$$

However,

zero covariance means uncorrelated, but it does not imply independence in general.

3.4 Example 1

Given

$$Y = -2X + 3$$

Find

$$\text{Cov}(X, Y)$$

Step 1: Use covariance formula

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Step 2: Substitute expression for Y

$$Y = -2X + 3$$

Step 3: Compute XY

$$\begin{aligned} XY &= X(-2X + 3) \\ &= -2X^2 + 3X \end{aligned}$$

Step 4: Compute expectation of Y

$$\begin{aligned} E[Y] &= E[-2X + 3] \\ &= -2E[X] + 3 \end{aligned}$$

Step 5: Substitute into covariance formula

$$\text{Cov}(X, Y) = E[-2X^2 + 3X] - E[X](-2E[X] + 3)$$

Expanding:

$$\begin{aligned} &= -2E[X^2] + 3E[X] + 2(E[X])^2 - 3E[X] \\ &= -2E[X^2] + 2(E[X])^2 \end{aligned}$$

Step 6: Use variance identity

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

Thus

$$\text{Cov}(X, Y) = -2\text{Var}(X)$$

3.5 Pearson's Correlation Coefficient

Pearson's correlation coefficient measures the **effect of change in one variable when the other variable changes**.

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

or

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

Properties:

- Measures **strength and direction of linear relationship**
- **Normalized version of covariance**
- **Unit-free**

Range:

$$-1 \leq \rho_{XY} \leq 1$$

Interpretation:

- $\rho_{XY} > 0 \rightarrow$ positive linear trend
 - $\rho_{XY} < 0 \rightarrow$ negative linear trend
 - $\rho_{XY} = 0 \rightarrow$ no linear correlation
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3.6 Example 2

Joint PDF:

$$f_{XY}(x, y) = \begin{cases} \frac{xy}{96}, & 0 < x < 4, 1 < y < 5 \\ 0, & \text{otherwise} \end{cases}$$

Compute:

1. $E[X]$
 2. $E[Y]$
 3. $E[XY]$
 4. $E[2X + 3Y]$
 5. $\text{Var}(X)$
 6. $\text{Var}(Y)$
 7. $\text{Cov}(X, Y)$
 8. ρ_{XY}
-

Step 1: Compute $E[X]$

$$\begin{aligned} E[X] &= \int_0^4 x f_X(x) dx \\ &= \int_0^4 x \cdot \frac{x}{8} dx \\ &= \frac{1}{8} \int_0^4 x^2 dx \\ &= \frac{1}{8} \left[\frac{x^3}{3} \right]_0^4 \\ &= \frac{1}{8} \cdot \frac{64}{3} \\ &= \frac{8}{3} \end{aligned}$$

Thus

$$E[X] = \frac{8}{3}$$

Step 2: Compute $E[Y]$

$$\begin{aligned} E[Y] &= \int_1^5 y f_Y(y) dy \\ &= \int_1^5 y \cdot \frac{y}{12} dy \\ &= \frac{1}{12} \int_1^5 y^2 dy \\ &= \frac{1}{12} \left[\frac{y^3}{3} \right]_1^5 \\ &= \frac{31}{9} \end{aligned}$$

Thus

$$E[Y] = \frac{31}{9}$$

Step 3: Compute $E[XY]$

Since X and Y are independent,

$$\begin{aligned} E[XY] &= E[X]E[Y] \\ &= \frac{8}{3} \cdot \frac{31}{9} \\ &= \frac{248}{27} \end{aligned}$$

Step 4: Compute $E[2X + 3Y]$

Using linearity of expectation:

$$\begin{aligned} E[2X + 3Y] &= 2E[X] + 3E[Y] \\ &= 2\left(\frac{8}{3}\right) + 3\left(\frac{31}{9}\right) \\ &= \frac{16}{3} + \frac{31}{3} \end{aligned}$$

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$$= \frac{4}{3}$$

Step 5: Compute $Var(X)$

$$Var(X) = E[X^2] - (E[X])^2$$

First compute $E[X^2]$

$$\begin{aligned} E[X^2] &= \int_0^4 x^2 \frac{x}{8} dx \\ &= \frac{1}{8} \int_0^4 x^3 dx \\ &= \frac{1}{8} \left[\frac{x^4}{4} \right]_0^4 \\ &= 8 \end{aligned}$$

Then

$$\begin{aligned} Var(X) &= 8 - \left(\frac{8}{3}\right)^2 \\ &= 8 - \frac{64}{9} \\ &= \frac{8}{9} \end{aligned}$$

Step 6: Compute $Var(Y)$

$$Var(Y) = E[Y^2] - (E[Y])^2$$

First compute $E[Y^2]$

$$\begin{aligned} E[Y^2] &= \int_1^5 y^2 \frac{y}{12} dy \\ &= \frac{1}{12} \int_1^5 y^3 dy \end{aligned}$$

$$= \frac{1}{12} \left[\frac{y^4}{4} \right]_1^5$$

$$= 13$$

Thus

$$\text{Var}(Y) = 13 - \left(\frac{31}{9}\right)^2$$

$$= \frac{92}{81}$$

Step 7: Compute Covariance

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

$$= \frac{248}{27} - \frac{8}{3} \cdot \frac{31}{9}$$

$$= \frac{248}{27} - \frac{248}{27}$$

$$= 0$$

Thus

$$\text{Cov}(X, Y) = 0$$

Step 8: Compute Pearson Correlation

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

Since

$$\text{Cov}(X, Y) = 0$$

$$\rho_{XY} = 0$$

(All steps correspond exactly to the solution shown on pages 15–19.)

4. Jointly Gaussian Random Variables

4.1 Definition

A pair (X, Y) is **jointly Gaussian** if

- its **joint density** has the **bivariate Gaussian form**, and
 - the **marginals** are **Gaussian**.
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4.2 Algebraic Form

The **joint Gaussian PDF** is

$$f_{XY}(x, y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)} \left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right)\right]\right)$$

Where

- μ_X, μ_Y = means
 - σ_X, σ_Y = standard deviations
 - ρ = Pearson correlation coefficient
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4.3 Interpretation

Example from the lecture:

Relationship between

- X : Atmospheric CO₂ concentration (ppm)
- Y : Global surface temperature anomaly (°C)

Because both variables vary due to many random environmental processes such as

- solar radiation
- ocean circulation
- emissions
- weather variability

their joint behavior can often be approximated by a **Joint Gaussian Distribution**.

If you want, I can also convert this **exact scribe** into **clean LaTeX (exam-ready PDF format)** since your workflow prefers **LaTeX for PDF generation**.

